FIBER-GRAND: Exact Inverse-Alignment Decoding for One Deletion and Substitutions

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***Abstract*—A direct extension of noise-guessing decoding to synchronization channels can rank individual edit histories even though maximum-likelihood (ML) decoding ranks the aggregate probability of each candidate word. We study arbitrary binary block codes observed through one uniformly located deletion and independent hard-decision substitutions. First, an explicit two-codeword family shows a strict best-history/aggregate-ML reversal. We then introduce an exact inverse-alignment decoder that enumerates substitution shells, tests code membership, evaluates the complete alignment-summed likelihood only for discovered codewords, and stops with a certified unseen-candidate bound. A realization-dependent theorem bounds the stopping shell and yields a high-probability membership/query exponent no larger than $h_2(p_s)$. In 996 exact finite-length trials, the decoder had no correctness or censoring failure. At $n = 32$, it required median complete likelihood evaluation for only 1.5–13.5 codewords across five code/rate cells, avoiding factors of 6.99×10^5 to 4.98×10^6 relative to exhaustive codeword scoring. The claims are in a membership-oracle model and do not claim universal wall-clock optimality.**

I. Introduction

Guessing Random Additive Noise Decoding (GRAND) obtains ML decoding on additive discrete channels by ordering noise realizations and stopping at the first inverse image that belongs to the codebook [1]. Synchronization errors change this logic: multiple deletion alignments can produce the same candidate, so candidate likelihood is a sum over compatible histories. Selecting the first codeword reached by one highly likely history need not maximize that sum.

Prior synchronization decoders include drift-state trellises and watermark constructions [2], sequential convolutional decoding [3], and code constructions specialized to deletions with substitutions [4]. A GRAND-assisted modulation receiver has also treated insertion/deletion effects through padding [5]. Our setting is different: the code is not redesigned, access is only through a membership test, and the requested output is exact aggregate ML for a finite binary block code. Generic threshold aggregation [6] explains why threshold-style stopping itself is not a standalone novelty claim.

This paper makes four narrow contributions. First, it gives a strict family where best-history and aggregate-ML orderings reverse. Second, it gives a code-modular shell-certified exact decoder. Third, it proves realization-dependent and high-probability query bounds. Fourth, it separates candidate generation, membership calls, complete scoring, and wall-clock calibration in an exact multi-code campaign. The paper is restricted to exactly one uniform deletion and BSC substitutions; multiple edits, soft information, code design, and hardware are outside scope.

II. Channel and pathwise failure

Let $X = (X_1, \ldots, X_n) \in \{0,1\}^n$. Independent substitutions with probability $p_s < 1/2$ act on the symbols, and exactly one coordinate J is deleted uniformly. The observation $Y \in \{0,1\}^m$ has $m = n-1$. The nonempty code $\mathcal{C} \subseteq \{0,1\}^n$ has a uniform prior and is accessed through $\mathbf{1}\{x \in \mathcal{C}\}$. For candidate x and observation y, define the mismatch count under deletion alignment j by

$$d_j(x,y) = \sum_{i<j} \mathbf{1}\{x_i \neq y_i\} + \sum_{i>j} \mathbf{1}\{x_i \neq y_{i-1}\}. \quad (1)$$

Conditioning on J gives

$$W(y \mid x) = \frac{1}{n} \sum_{j=1}^{n} p_s^{d_j(x,y)} (1-p_s)^{m-d_j(x,y)}. \quad (2)$$

All d_j are obtained in $O(n)$ arithmetic operations from

$$d_{j+1} = d_j - \mathbf{1}\{x_{j+1} \neq y_j\} + \mathbf{1}\{x_j \neq y_j\}. \quad (3)$$

Theorem 1 (Strict best-history/aggregate-ML reversal)**.** *For $n \geq 4$, $0 < p_s < 1/2$, let*

$$y = 0^{n-3}10, \quad x_A = 0^{n-3}101, \quad x_B = 0^n, \quad \mathcal{C} = \{x_A, x_B\}.$$

The most likely individual compatible history strictly favors x_A, while aggregate ML favors x_B if and only if

$$np_s > 1 + 2p_s^2. \quad (4)$$

Proof sketch. Deleting the final symbol of x_A requires no substitution, while every alignment of x_B requires one. The mismatch multiplicities of x_A at distances $0,1,2,3$ are $1,1,1,n-3$. Substitution in (2) and factorization give

$$\frac{n(W(y \mid x_B) - W(y \mid x_A))}{(1-p_s)^{n-4}} = (1-2p_s)(np_s - 1 - 2p_s^2),$$

which establishes both strict statements. Full proofs are in the companion supplement. □

Algorithm 1 Baseline one-deletion FIBER-GRAND

$\mathsf{seen} \leftarrow \emptyset$; no incumbent
for $s = 0, 1, \ldots, m$ **do**
  for all $e \in \{0,1\}^m$ with $\mathrm{wt}(e) = s$ **do**
    $z \leftarrow y \oplus e$
    for all $x \in \mathcal{I}(z)$ using the $n+1$ distinct generator **do**
      if $x \notin \mathsf{seen}$ **then**
        insert x into seen and query $x \in \mathcal{C}$
        if $x \in \mathcal{C}$ **then**
          evaluate the complete score (2) via (3)
          update the incumbent score and complete tie set
  if an incumbent exists and [$s = m$ or incumbent $> U_s$] **then**
    return the incumbent and complete tie set

III. Exact inverse-alignment decoder

For $z \in \{0,1\}^m$, let $\mathcal{I}(z)$ denote all binary length-n supersequences obtained by one insertion. Its classical cardinality is $n+1$: for inserted bit b, it suffices to insert in the first gap and immediately after each symbol unequal to b. Define the discovery shell

$$h_y(x) = \min_j d_j(x, y).$$

Candidate x is generated by the end of shell s exactly when $h_y(x) \leq s$.

Proposition 2 (Exact discovery shell)**.** *Candidate x first appears in the inverse-alignment enumeration at shell $h_y(x)$.*

Proof. For any alignment j, deleting x_j gives a length-m word z for which $z = y \oplus e$ and $\mathrm{wt}(e) = d_j(x, y)$; reinserting x_j at coordinate j reconstructs x. Conversely, every generated word comes with such an alignment and error vector. Minimizing over j proves both the first-appearance and end-of-shell statements. □

After shell $s < m$, an unseen candidate satisfies $d_j \geq s+1$ for every alignment, hence

$$U_s = p_s^{s+1}(1-p_s)^{m-s-1} \tag{5}$$

upper-bounds its complete likelihood.

Theorem 3 (Exact shell certificate)**.** *After shell $s < m$, every unseen word has likelihood at most U_s. If a completely scored incumbent has likelihood strictly greater than U_s, the incumbent set is the complete global ML tie set. Algorithm 1 always terminates and returns exact ML.*

Proof. Proposition 2 implies $d_j \geq s+1$ for every alignment of an unseen word. Because $p_s/(1-p_s) < 1$, each component in (2) is at most U_s, and their uniform average is no larger. Strict comparison excludes both a better unseen codeword and an unseen tie. At $s = m$, all binary length-n words have been generated, so finite exhaustion completes the proof. □

The strict inequality is required to certify the complete tie set. This is an exact decoder, not an abandonment rule. At $p_s = 0$, shell zero contains every positive-likelihood word; at $p_s = 1/2$, shell ordering loses strict decay and only finite exhaustion remains.

IV. Stopping and query complexity

Let T be the realized number of substitutions among the m surviving symbols, set $\rho = p_s/(1-p_s)$, and define

$$L_n(p_s) = \min\{\ell \geq 1 : \rho^\ell < 1/n\} = \left\lfloor \log_{(1-p_s)/p_s} n \right\rfloor + 1. \tag{6}$$

Theorem 4 (Realization-dependent stopping)**.** *Algorithm 1 stops no later than*

$$S \leq \min\{m, T + L_n(p_s) - 1\}. \tag{7}$$

Consequently, with $\mathrm{Vol}(m, s) = \sum_{w=0}^{\min\{m,s\}} \binom{m}{w}$,

$$Q_{\mathrm{gen}}, Q_{\mathrm{mem}} \leq (n+1)\,\mathrm{Vol}(m, S), \qquad Q_{\mathrm{score}} \leq Q_{\mathrm{mem}}. \tag{8}$$

Proof sketch. The transmitted word is discovered by shell T and contributes at least $\frac{1}{n} p_s^T (1-p_s)^{m-T}$. At $s = T + L_n - 1$, the ratio of (5) to this component is $n\rho^{L_n} < 1$, so the strict certificate holds. Counting the $n+1$ distinct supersequences per error vector gives (8). □

Corollary 5 (High-probability membership/query exponent)**.** *For fixed $0 < p_s < 1/2$ and any $c > 0$, with probability at least $1 - n^{-c}$,*

$$Q_{\mathrm{mem}}, Q_{\mathrm{score}} \leq 2^{n(h_2(p_s)+o(1))}.$$

For a code sequence of rate $R > h_2(p_s)$, exhaustive codeword-wise ML therefore performs at least $2^{n(R-h_2(p_s)-o(1))}$ times as many complete candidate-likelihood evaluations.

The proof applies Hoeffding concentration to T and a Hamming-ball volume bound. More explicitly, for $\epsilon_n = \sqrt{c \ln n/(2m)}$, Hoeffding gives $\Pr[T > m(p_s + \epsilon_n)] \leq n^{-c}$, while $L_n = O(\log n)$. Thus $S/m \leq p_s + o(1) < 1/2$ on the complementary event and $\mathrm{Vol}(m, S) \leq 2^{m h_2(S/m)}$. The polynomial factor $n+1$ is absorbed in $2^{o(n)}$. This statement is in a membership-oracle model. It does not assign unit cost to every membership implementation or claim superiority to every code-specific exact decoder.

For finite confidence, let τ_δ be the smallest binomial quantile satisfying $\Pr[T \leq \tau_\delta] \geq 1 - \delta$. Then, with probability at least $1 - \delta$,

$$Q_{\mathrm{mem}} \leq (n+1)\,\mathrm{Vol}(m, \min\{m, \tau_\delta + L_n - 1\}). \tag{9}$$

Table I evaluates this conservative cap at the simulated $p_s = 0.05$. The caps are deliberately code-independent and can be loose; the measured median $n = 32$ membership count is below 10^3 in every cell.

TABLE I
FINITE-CONFIDENCE THEOREM CAPS AT $p_s = 0.05$.

n	δ	L_n	τ_δ	$(n+1)\,\mathrm{Vol}(n-1, \tau_\delta + L_n - 1)$
16	10^{-2}	1	3	9,792
16	10^{-3}	1	4	32,997
24	10^{-2}	2	4	1,113,800
24	10^{-3}	2	5	3,637,475
32	10^{-2}	2	5	31,107,417
32	10^{-3}	2	6	117,883,392

TABLE II
FROZEN PHASE-2C EXPERIMENT DESIGN.

Item	Contract
Channel	one uniform deletion; BSC $p_s = 0.05$
Blocklengths	$n \in \{16, 24, 32\}$
Rates/codes	approx. 2/3, 3/4; CRC, random linear, Hamming
Natural trials	896 total; 64 per cell
Stress trials	100 $n = 32$, surviving weights 0–4
Exact comparators	history-pair, exhaustive, completed B&B
Arithmetic	checked 512-bit integers

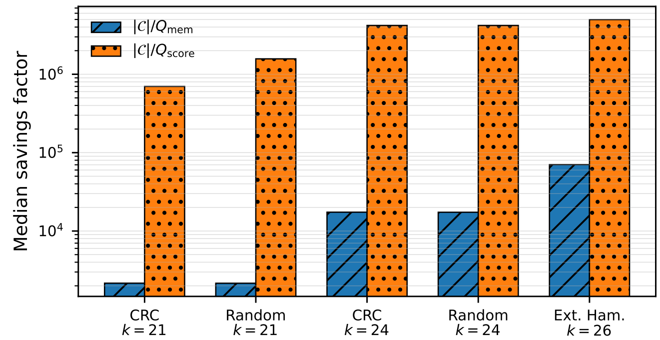

Fig. 1. Frozen $n = 32$ median savings relative to codebook-wise work. Complete likelihood scoring is reduced far more than membership queries because only discovered codewords are scored.

V. Exact finite-length evidence

The implementation uses 512-bit exact integer scores and no floating-point stopping comparison. Table II records the frozen preregistered campaign. Random-linear cells rotate over four independently generated code instances. The generic branch-and-bound baseline searches message coefficients with a valid likelihood upper bound and is included only when it completes within the frozen node/time cap.

The campaign contained 996 trials. Exact theorem-support validation covered 335146 cases. Every shell decode and history-pair decode completed; there were zero validation violations. All 384 exhaustive and 432 completed generic branch-and-bound comparisons returned the same ML score and complete tie set.

At $n = 32$, the median decoder stopped at shell one in all five cells. It tested roughly 10^3 distinct words for membership but completely scored only 1.5–13.5 codewords. The score-saving factor was at least 1.91×10^5 at the 10th percentile in every $k = 21$ cell and at least 1.86×10^6 in the $k \geq 24$ cells. On the 16 preregistered $k = 21$ exhaustive calibrations, the median exhaustive/FIBER wall-clock ratio was 465.6. This is a finite implementation comparison, not an instance-optimality claim.

A. What the savings do and do not show

The distinct insertion-sphere generator attempted about 1.94× fewer hypotheses than the $2n$ history-pair generator, but its wall-clock advantage did not pass the preregistered gate. Thus the classical $n+1$ generation result is retained as a clean implementation device, not a latency contribution. Against the generic branch-and-bound implementation, FIBER was faster in four of five $n = 32$ cells and slower in several $n = 24$ cells. The strongest practical statement is the large reduction relative to exhaustive codeword-wise scoring and the competitive finite behavior against this one generic exact search; no universal latency claim is made.

The sampled high-rate cells have high ML block-error rates. This does not indicate decoder failure: all exact comparators agree on the ML result. It does limit system-performance interpretation, and this paper therefore makes a decoder-complexity claim rather than a coding-gain claim.

B. Best history versus aggregate ML

The campaign recorded many cases in which a lexicographically selected first-shell representative differed from the selected aggregate-ML representative, but the corresponding tie sets always overlapped. The finite campaign therefore had no case where the first-shell codeword tie set was disjoint from the aggregate-ML tie set. These representative differences are not used as evidence of strict pathwise suboptimality; Theorem 1 supplies a strict structural family. A larger prevalence study would answer a different question and is not needed for exactness.

VI. Claim boundaries and conclusion

The generic ideas of summing paths, threshold aggregation, membership-oracle decoding, and the $n+1$ insertion-sphere cardinality are not claimed as new. The defensible contributions are the strict channel/coding family, the exact shell-certified specialization, the stop/query theorem, and the exact evidence separating membership, scoring, generation, and time. For a one-deletion-plus-substitution channel, ranking histories and ranking candidate words are different operations. FIBER-GRAND restores exact aggregate ML while retaining code membership as the code interface. The stopping theorem gives a channel-specific query bound, and exact finite-length evidence shows repeated reductions of several orders of magnitude in complete codeword scoring. The current evidence supports a narrow Paper-I contribution. Before any submission claim, the proof and novelty positioning require external proof and novelty review; no broader synchronization-channel claim follows from this model.

TABLE III

FROZEN $n = 32$, $p_s = 0.05$ NATURAL-CHANNEL EVIDENCE (64 TRIALS PER CELL). RATIOS ARE MEDIANS UNLESS MARKED OTHERWISE. A DASH MEANS EXHAUSTIVE CALIBRATION WAS NOT RUN AT THAT DIMENSION.

Code / dimension	Q_{mem}	Q_{score}	$\lvert\mathcal{C}\rvert/Q_{\text{score}}$	P10 $\lvert\mathcal{C}\rvert/Q_{\text{score}}$	$\lvert\mathcal{C}\rvert/Q_{\text{mem}}$	Exhaustive / FIBER	B&B / FIBER
CRC, $k = 21$	970.5	3	699051	190650	2160.9	522.0	0.991
Random, $k = 21$	972.0	1.5	1.57×10^6	262144	2157.6	465.6	1.66
CRC, $k = 24$	964.5	4	4.19×10^6	2.1×10^6	17395	–	1.2
Random, $k = 24$	966.5	4	4.19×10^6	1.86×10^6	17359	–	2.39
Ext. Hamming, $k = 26$	954.0	13.5	4.98×10^6	3.36×10^6	70345	–	2.18

REPRODUCIBILITY STATEMENT

All reported values are generated from immutable commit `2f7736b70bf7f6f659a6707fad716407635a7deb`. The campaign specification, exact C++ source, trial-level outputs, theorem checks, and manifests are retained in the public research repository. Phase 2D introduces no new simulation.

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